

PREDICTIVE LEARNING SYSTEM FOR MATHEMATICS FORM 4 AND FORM 5

FYP 1 | BCS

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BACKGROUND

The Fourth Industrial Revolution represents a significant change in the modern workforce. Malaysia's education sector is shifting towards the use of an intelligent system. Mathematics being one of the core subjects in SPM, predicting students with high comprehension and mastery level is critical.



OBJECTIVES

- To develop a predictive learning system that classifies students' mastery levels
- To develop an adaptive quiz system that dynamically adjusts question difficulty
- To develop an intelligent guidance chatbot module



COMMERCIAL POTENTIAL

- Educational Institution**
Can be deployed in schools and tuition centers
- Educators**
Integrate AI driven learning ecosystem into classroom

PROBLEM STATEMENT

- Conventional study methods do not accommodate for the students' proficiency level.
- Lack of a dedicated digital learning platform that solely focuses on the current Mathematics KSSM Form 4 and Form 5 syllabus.
- Absence of a real time guidance tool that can help students when they are stuck on a hard question.

22%
Failure rate of students

76%
Academic Impact

23%
Score Improvement

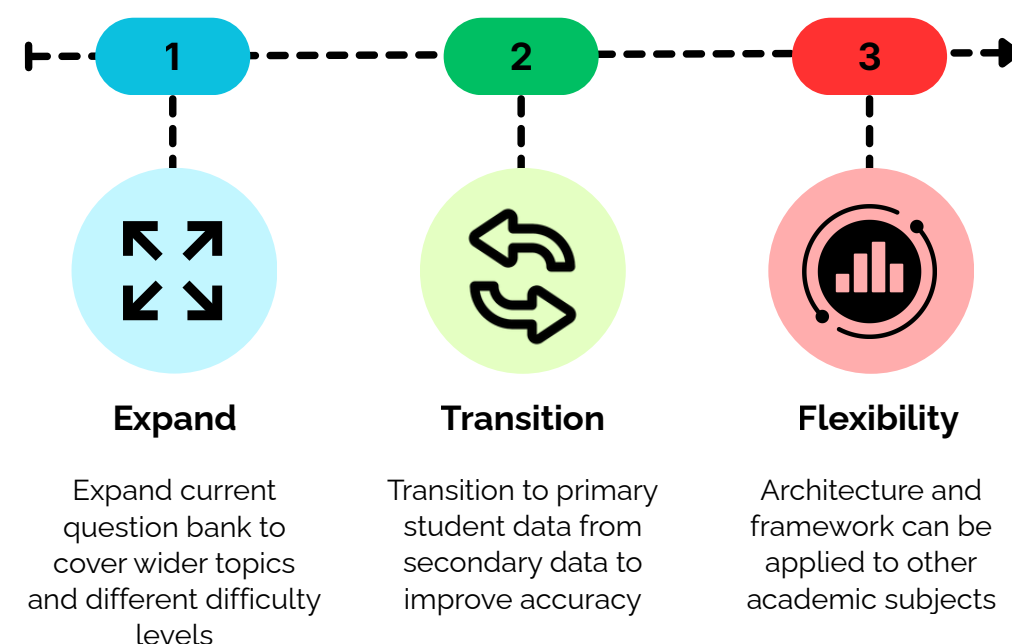
SOLUTION

A Predictive Learning System. By analyzing student's behavioral and academic data, the system can provide personalized learning through the integration of machine learning techniques

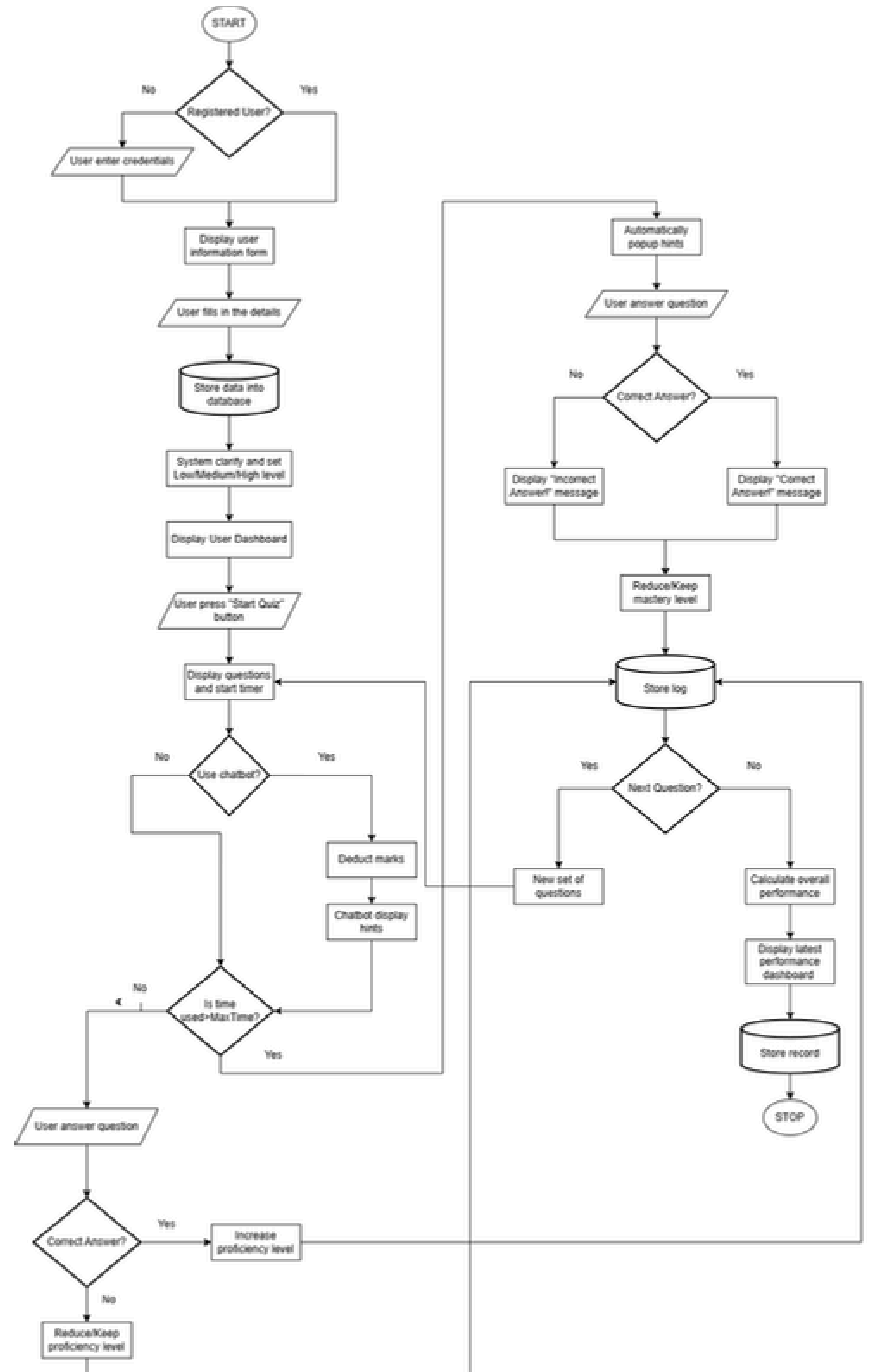
TECHNOLOGIES



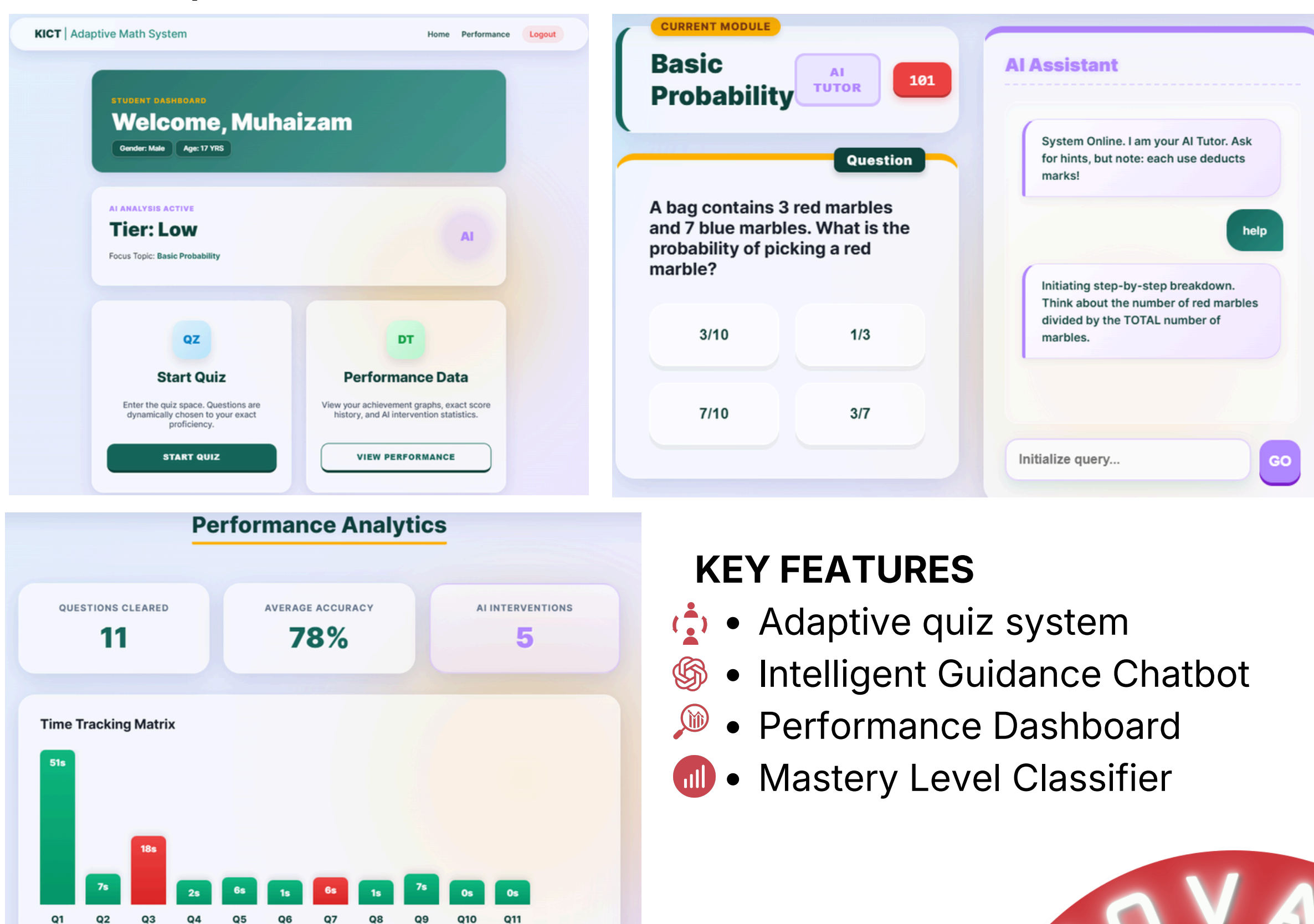
FUTURE PLANS



FLOW CHART



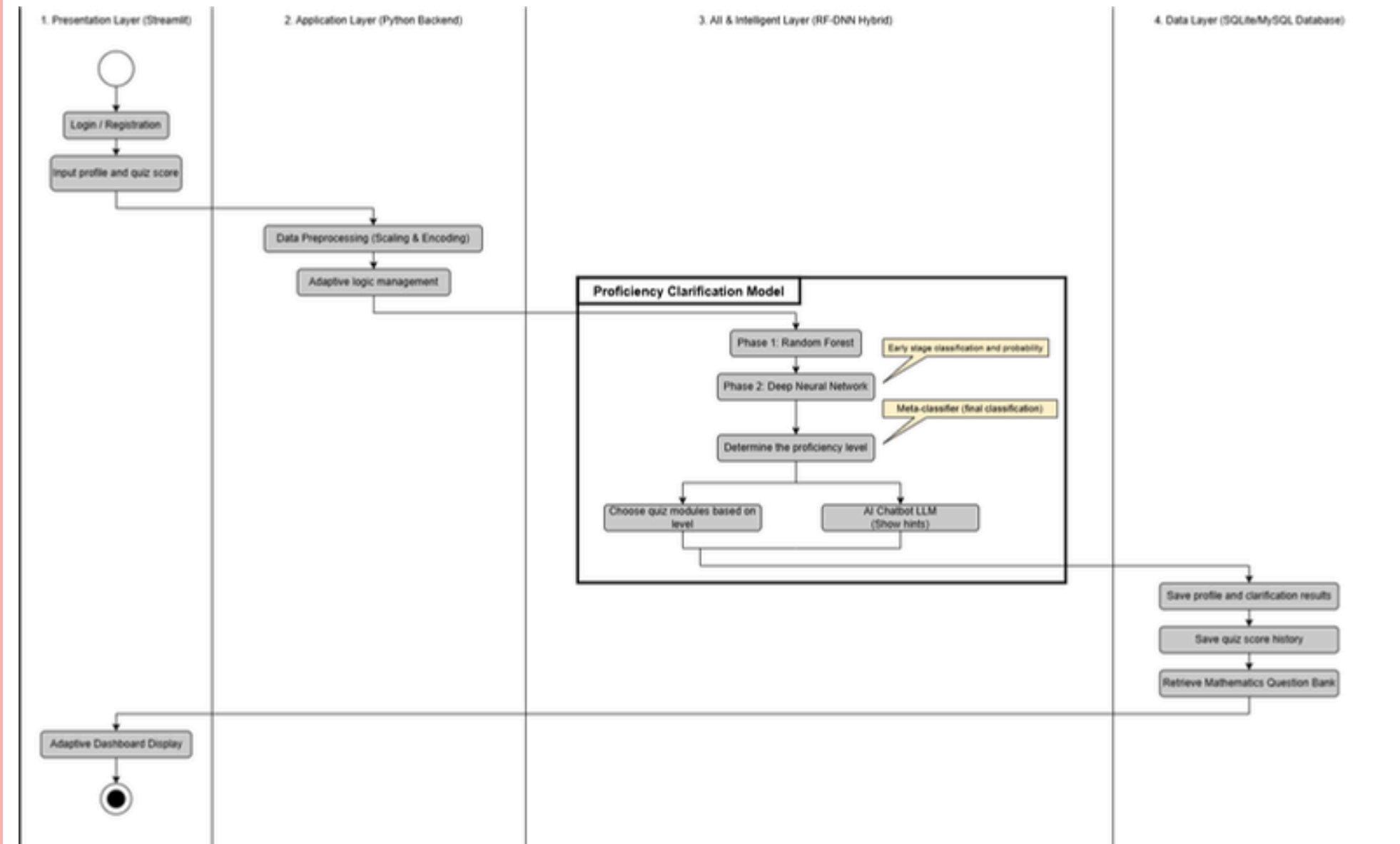
PRODUCT/PROTOTYPE



KEY FEATURES

- Adaptive quiz system
- Intelligent Guidance Chatbot
- Performance Dashboard
- Mastery Level Classifier

ARCHITECTURE DIAGRAM



COMMUNITY IMPACTS

- Form 4 and Form 5 Students**
 - Enables early detection of learning gaps
 - Provides personalized pathways for effective revision
- Teachers**
 - Provides a data-driven dashboard to track class progress
 - Identify weak students early for allowing timely interventions